

BACKGROUND OF THE INVENTION

Accurate and repeatable measurement of penile anatomical geometry presents substantial technical challenges when performed using conventional consumer or clinical approaches. Traditional measurement techniques such as rigid rulers, flexible measuring tapes, printed sizing guides, or visual estimation rely heavily on user interpretation and manual positioning. These techniques are prone to inconsistent anatomical reference points, variable applied pressure, deformation of soft tissue, measurement angle deviation, and non-standardized physiological states. As a result, conventional methods frequently produce inconsistent or non-reproducible measurements.

Penile anatomy exhibits significant inter-individual variability in longitudinal length, cross-sectional radius distribution, curvature, taper characteristics, and localized asymmetry. In addition, intra-individual variability may occur depending on vascular state, arousal level, temperature, hydration, and environmental conditions. Existing tools are not designed to account for motion artifacts, orientation drift, or three-dimensional surface reconstruction. Most known approaches generate only one or two scalar outputs (for example, length or circumference), without capturing meaningful structural distributions along the longitudinal axis.

The absence of standardized digital geometry profiling limits the ability to perform precise device calibration, structural fitment optimization, compatibility modeling, longitudinal monitoring, or population-level structural analysis. There remains a need for a sensing architecture capable of capturing spatial surface data in a controlled manner and transforming such data into a structured digital geometry representation.

There further exists a need for systems that combine time-of-flight distance ranging with optical motion sensing in order to reconstruct anatomical geometry through sequential spatial sampling along a defined measurement path. By integrating distance measurements with motion tracking and computational reconstruction, it becomes possible to generate a repeatable, privacy-preserving, user-specific penile geometry profile that is suitable for digital storage, algorithmic analysis, temporal comparison, and integration with broader biometric or personalization systems.

The present disclosure addresses these technical deficiencies by providing systems and methods that capture radial distance data relative to a reference frame, detect controlled longitudinal displacement during scanning, compensate for angular deviation where necessary, and reconstruct a structured three-dimensional geometric model from the resulting dataset.